



PROJECT GUIDE AUR PRESENTATION NOTES

SireSoft Scan Tool

Kaise kaam karta hai, kaise chalta hai, data kahan hai

Server, port, commands, database, history, edit ki hui receipts, faide aur kamzoriyan - project pesh karne ke liye sab kuch.

Server	lsnet - 10.0.2.2 (OpenVPN par)
Address	http://10.0.2.2:9001
Project folder	/home/siresoft/siresoft-receiptiq
Database	PostgreSQL - siresoft_receiptiq_db
AI model	Ollama qwen2.5v1:3b, server par hi chalta hai
Code	github.com/Engr-Mahnoor-code/siresoft-receiptiq (private)

SireSoft - andruni document - October 2026

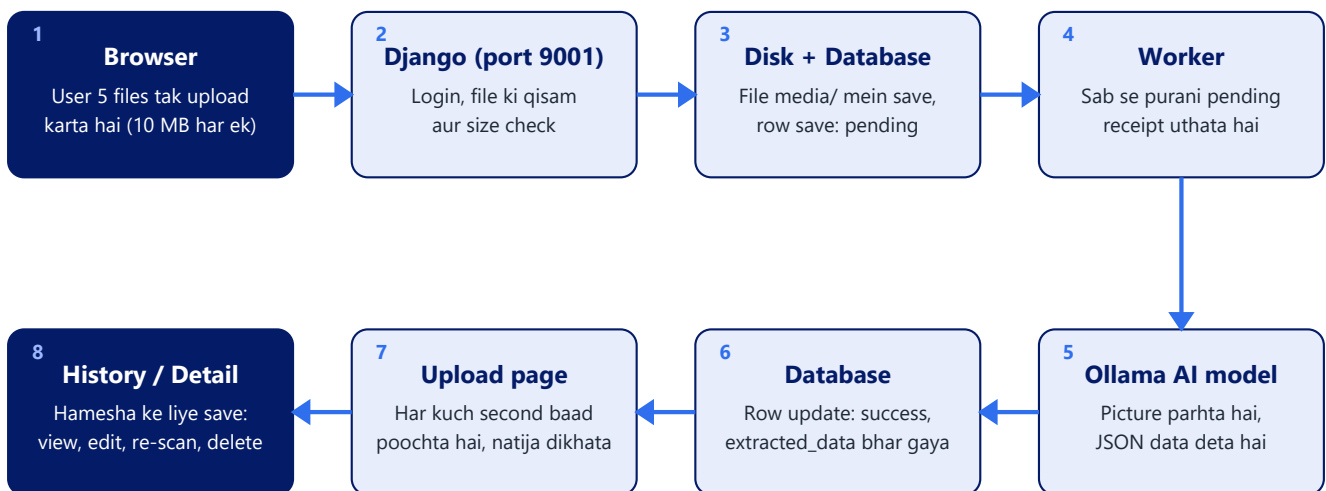
01 Project kya hai

SireSoft Scan Tool company ki andruni web application hai jo kisi receipt ya invoice ki photo ya PDF ko saaf, tarteeb-wala data bana deti hai. User browser mein receipt upload karta hai; SireSoft ke apne server par chalne wala AI vision model usay parhta hai; aur natija - dukaan ka naam, address, tareekh, har item, subtotal, tax, doosri charges, grand total aur chhoti insights - PostgreSQL database mein save ho kar screen par dikhta hai.

Har receipt mehfooz rehti hai: original file, AI ne kya parha, aur baad mein kisi ne jo correction ki - kis ne aur kab ki, sab ke saath. Koi receipt bahar ki kisi AI service ko nahi jaati - model company ke server par hi chalta hai.

Hissa	Technology	Kaam
Front end	HTML, CSS, JavaScript (Django templates)	Login, Upload, History, Receipt detail aur Edit ke pages
Back end	Python 3 + Django 6 + Django REST Framework	Pages, upload API, access ke qaide, edit, delete
Web server	Gunicorn (3 workers), systemd ke neeche	App ko port 9001 par chalata hai, crash ho to khud restart
AI reader	Ollama + qwen2.5v1:3b vision model	Receipt ki picture parh kar JSON deta hai. Server par hi chalta hai
Background worker	Django command <code>process_receipts</code>	Line mein lagi receipts ek ek kar ke AI ko bhejta hai
Database	PostgreSQL - <code>siresoft_receiptiq_db</code>	Users aur receipts (table <code>receipts_receipt</code>)
File storage	Server ki disk - <code>media/receipts/</code>	Original uploads aur corrected copy pictures

02 Ek receipt system mein kaise safar karti hai



- 1. Upload.** User 5 files tak chunta hai (JPG, PNG, WEBP, GIF ya PDF, har ek 10 MB tak) aur Analyze dabata hai.

2. **Foran save.** Django har file disk par `media/receipts/` mein rakhta hai aur `receipts_receipt` table mein ek row banata hai jiska status `pending` hota hai. Browser ko foran jawab mil jata hai - wo AI ka intezar nahi karta.
3. **Worker uthata hai.** Background worker (apni alag systemd service) row ko `processing` karta hai aur picture local Ollama model ko bhejta hai. PDF pehle picture mein badli jaati hai (page 1).
4. **AI parhta hai.** Model JSON wapas deta hai: dukaan, items, bill summary, doosri charges aur insights. Code numbers aur text ko saaf karta hai.
5. **Natija save.** Row `success` ho jaati hai; dukaan ka naam, tareekh aur grand total apne columns mein aa jate hain, aur poora data `extracted_data` mein (aur ek pakki copy `scanned_data` mein).
6. **Live dikhta hai.** Upload page har kuch second baad server se poochta hai aur har natija tayyar hote hi dikha deta hai. User browser band bhi kar de to kaam server par chalta rehta hai.
7. **History.** Receipt ab History page par aa jaati hai, jahan usay dekha, edit, re-scan ya delete kiya ja sakta hai.

Agar AI koi receipt na parh sake to row `failed` ho jaati hai aur user ko saaf message milta hai. Agar Ollama hi band ho to receipt zaya nahi hoti: wapas line mein chali jaati hai aur khud dobara try hoti hai (3 baar tak).

03 Kya project chalane ke liye koi command chalani padti hai?

Nahi. Project server par har waqt pehle se chal raha hai. OpenVPN connect karein, browser mein `http://10.0.2.2:9001` kholein, aur project chal jayega - server mein login kiye baghair aur koi command chalaye baghair.

Wajah ye hai ke app server (`lsnet`, 10.0.2.2) par **systemd services** ki shakal mein install hai. Server on hote hi systemd inhe khud start karta hai, aur agar kabhi ruk jayein to khud restart kar deta hai. Laptop band karna, SSH band karna ya VPN disconnect karna app ko band nahi karta - app server par chalti hai, aap ke computer par nahi.

Service	Kya hai	Kis liye zaroori
<code>siresoft-receptiq</code>	Website (Gunicorn, port 9001)	Koi bhi page kholne ke liye
<code>siresoft-receptiq-worker</code>	Background reader	Receipt ka Queued se Done hona
<code>postgresql</code>	Database server	Sab kuch save karna
<code>ollama</code>	Local AI model server (localhost:11434)	Receipts parhna
<code>siresoft-receptiq-tunnel</code>	Cloudflare tunnel (optional)	VPN ke baghair access

Kaunsa address kab chalta hai

Soorat-e-haal	Address	Chalega?
OpenVPN connected	http://10.0.2.2:9001	Haan
OpenVPN disconnected	http://10.0.2.2:9001	Nahi - 10.0.2.2 private address hai jo sirf VPN ke andar hota hai
OpenVPN disconnected	Cloudflare tunnel ka address (https://...trycloudflare.com)	Haan, agar tunnel service chal rahi ho
Server band / restart ho raha	koi bhi	Nahi - server wapas aane ka intezaar karein; services phir khud start ho jati hain

App port 9001 par is liye chalti hai kyunke server ki `.env` file (`/home/siresoft/siresoft-receptiq/.env`) mein `APP_PORT=9001` likha hai. Is file mein saari settings aur passwords hain, aur ye kabhi GitHub par upload nahi hoti.

Server par `python manage.py runserver mat` chalayein. Wo development server hai; wo waise bhi mana kar dega kyunke Gunicorn pehle se port 9001 par hai. Asal app pehle hi chal rahi hai.

04 Server ki commands

Saari commands aap ke laptop ke terminal (Git Bash ya PowerShell) mein likhi jati hain jab VPN connected ho. `#` se shuru hone wala hissa sirf wazahat hai, command nahi. `sudo` wali commands ek waqt mein ek chalayein - `sudo` password maang sakta hai.

1. Server mein login

```
# laptop se, VPN on ho; server ka password daalein
ssh siresoft@10.0.2.2

# project ke folder mein jayein
cd ~/siresoft-receptiq
```

2. Check karein ke sab chal raha hai

```
# app + worker: do lines, dono mein "active" aana chahiye
systemctl is-active siresoft-receptiq siresoft-receptiq-worker

# jawab HTTP/1.1 302 Found aana chahiye
curl -I http://10.0.2.2:9001/

# zyada tafseel (bahar aane ke liye q dabayein)
systemctl status siresoft-receptiq --no-pager
```

Sahi natija: har line mein `active`, aur curl se `HTTP/1.1 302 Found` (302 ka matlab hai app logged-out visitor ko login page par bhej rahi hai - ye bilkul theek hai).

3. Start, stop, restart

```
# app aur worker restart
sudo systemctl restart siresoft-receiptiq siresoft-receiptiq-worker

# band karna (website offline ho jayegi!)
sudo systemctl stop siresoft-receiptiq siresoft-receiptiq-worker

# dobara chalana
sudo systemctl start siresoft-receiptiq siresoft-receiptiq-worker
```

4. Kya ho raha hai, dekhein (logs)

```
# worker live - dekhna band karne ke liye Ctrl+C
journalctl -u siresoft-receiptiq-worker -f

# website ki aakhri 50 lines
journalctl -u siresoft-receiptiq -n 50 --no-pager

# sirf errors
journalctl -u siresoft-receiptiq -p err -n 30 --no-pager
```

Presentation mein live dikhane ke liye worker ka log sab se acha hai: har receipt par Receipt #84 (file.jpg) - extracting... aur phir Receipt #84 done in 41.2s - Dukaan ka naam, total 26.06 aata hai.

5. Naya code server par lagana (GitHub par push ke baad)

```
cd ~/siresoft-receiptiq

# GitHub se naya code lein
git pull

# is server par har pull ke baad zaroori
sudo restorecon -v deploy/*.sh

# sirf tab jab database ka dhaancha badla ho
./venv/bin/python manage.py migrate

# nayi CSS/JS copy - warna purana design rehta hai
./venv/bin/python manage.py collectstatic --noinput

# naya code load karein
sudo systemctl restart siresoft-receiptiq siresoft-receiptiq-worker
```

6. VPN band ho to public address

```
# Cloudflare ka maujooda address (har restart par badalta hai)
journalctl -u siresoft-receiptiq-tunnel \
  | grep -o 'https://[a-z0-9-]*\.trycloudflare\.com' | tail -1
```

05 Data kahan save hota hai

Do jagah: **database** mein maloomat, aur **server ki disk** par files.

Database: PostgreSQL, database siresoft_receiptiq_db

Table	Ek row kya hai
receipts_receipt	Ek upload ki hui receipt - columns neeche
auth_user	Ek user account (username, email, staff/admin, password hash ki shakal mein)
django_session	Is waqt kaun logged in hai

receipts_receipt **ke columns**

Column	Matlab
id	Receipt ka number - wahi jo address mein dikhta hai, maslan /receipt/ 84 /
user_id	Kis ne upload ki (auth_user.id se juda)
vendor_name, date, total_amount	Dukaan ka naam, receipt ki tareekh aur grand total - History list inhi se banti hai
file	Original upload ka path, maslan receipts/bill_01.jpg
content_type	File ki qisam, maslan image/jpeg ya application/pdf
status	pending -> processing -> success ya failed
extracted_data	Poora data JSON mein: dukaan, details, items, bill summary, doosri charges, insights. Edit ke baad isme corrected values hoti hain
scanned_data	AI ne shuru mein jo parha. Edit se kabhi nahi badalta - taake pehle aur baad ka muqabla ho sake
corrected_file	Corrected copy picture ka path, maslan receipts/corrected/receipt-84-corrected.png (sirf edit ke baad)
edited_at, edited_by_id	Kab aur kis ne receipt correct ki
created_at, started_at, processed_at	Upload hui, AI ne shuru kiya, AI ne khatam kiya
error_message, attempts	Fail hui to kyun, aur kitni baar dobara try hui

Disk par files

Server par folder	Andar kya hai
~/siresoft-receiptiq/media/receipts/	Har original upload, bilkul waisi jaisi user ne bheji. Kabhi nahi badalti
~/siresoft-receiptiq/media/receipts/corrected/	Har edit ki hui receipt ki ek PNG: SireSoft design mein saaf copy, corrected values ke saath

Files public nahi hain. Ye sirf app ke zariye milti hain (/receipt/84/file/, /receipt/84/corrected/), jo pehle check karti hai ke dekhne wala malik hai ya admin. Receipt delete karne se uski row **aur** dono files hat jati hain.

06 Saboot: receipts sach mein database mein save ho rahi hain

Presentation ka asal saboot yahi hai: receipt upload karein aur usay database mein aate hue dikhayein. Teen tareeqe, aasan se technical tak.

Tareeqa 1 - Browser mein Admin panel (koi command nahi)

1. **http://10.0.2.2:9001/admin/** kholein aur admin ke **username** se login karein (aam login page email leta hai; /admin/ username leta hai).
2. **Receipts** par click karein. Database ki har receipt nazar aayegi: id, dukaan, tareekh, total, status, user, upload ka waqt. Daayein taraf status ya user se filter karein.
3. Kisi ek receipt par click karein to uska sab kuch dikhega, poora extracted JSON aur disk par file kahan rakhi hai, samet.
4. Admin view jaan boojh kar sirf parhne ke liye (read-only) hai, taake ghalti se kuch na badle.

Tareeqa 2 - Seedha PostgreSQL (psql)

Pehle server mein login karein (section 4), phir database kholein:

```
# database kholein (sudo maange to server password)
sudo -u postgres psql -d siresoft_receiptiq_db

-- lambi rows ke liye har field alag line mein (optional)
\x auto
-- kaam khatam ho to psql se bahar
\q
```

Ab in mein se koi bhi query likhein. Har query ; par khatam hoti hai.

Kitni receipts, har status ki:

```
SELECT status, COUNT(*) FROM receipts_receipt GROUP BY status;
```

10 sab se nayi receipts:

```
SELECT id, vendor_name, date, total_amount, status, created_at
FROM receipts_receipt
ORDER BY id DESC
LIMIT 10;
```

Kis ne kya upload kiya:

```
SELECT r.id, u.username, u.email, r.vendor_name, r.total_amount, r.created_at
FROM receipts_receipt r
JOIN auth_user u ON u.id = r.user_id
ORDER BY r.id DESC
LIMIT 20;
```

Jo receipts fail huin, aur kyun:

```
SELECT id, created_at, error_message
FROM receipts_receipt
WHERE status = 'failed'
ORDER BY id DESC;
```

Tareeqa 3 - Django shell (database password ki zaroorat nahi)

```
cd ~/siresoft-receiptiq

# receipts ginein aur nayi paanch dikhayein
./venv/bin/python manage.py shell -c "from apps.receipts.models import Receipt as R; \
    print(R.objects.count()); \
    [print(r.id, r.status, r.vendor_name, r.total_amount) for r in R.objects.all()[:5]]"

# ek receipt: status, total aur file kahan hai
./venv/bin/python manage.py shell -c "from apps.receipts.models import Receipt as R; \
    r = R.objects.get(pk=84); print(r.status, r.total_amount, r.file.path)"
```

07 Ek individual receipt dekhna

Har receipt ka ek number (id) hota hai. Wo browser ke address mein hota hai:

<http://10.0.2.2:9001/receipt/84/> matlab receipt **84**. Neeche ki commands mein 84 ki jagah apna number likhein.

App mein

- History -> **View** se receipt khulti hai: dukaan, details, items, bill summary (grand total se pehle insights ke saath) aur insights.
- Neeche picture: **Original scan** tab mein file bilkul waisi jaisi upload hui; edit ke baad **Corrected copy** tab bhi aa jata hai.

Database mein

Receipt 84 ke main columns:

```
SELECT id, vendor_name, date, total_amount, status, content_type,
       file, corrected_file, created_at, processed_at, edited_at
FROM receipts_receipt
WHERE id = 84;
```

Al ne jo parha (ya corrected values), saaf format mein:

```
SELECT jsonb_pretty(extracted_data) FROM receipts_receipt WHERE id = 84;
```

Sirf items, table ki shakal mein:

```
SELECT item->>'name' AS item, item->>'quantity' AS qty,
       item->>'unit_price' AS unit_price, item->>'total' AS total
FROM receipts_receipt, jsonb_array_elements(extracted_data->'items') AS item
WHERE id = 84;
```


Disk par uski files

```
# saari original uploads, nayi pehle
ls -lht ~/siresoft-receiptiq/media/receipts/ | head -20

# saari corrected copy pictures
ls -lht ~/siresoft-receiptiq/media/receipts/corrected/

# receipt 84 ki file ka path, database se
sudo -u postgres psql -d siresoft_receiptiq_db \
  -c "SELECT file, corrected_file FROM receipts_receipt WHERE id = 84;"
```

Presentation ke liye file apne laptop par laane ke liye ye command apne **laptop** par chalayein (server par nahi):

```
# receipt 84 ki corrected picture is folder mein copy
scp siresoft@10.0.2.2:~/siresoft-receiptiq/media/receipts/corrected/receipt-84-corrected.png .

# ek original upload copy (naam file column se lein)
scp siresoft@10.0.2.2:~/siresoft-receiptiq/media/receipts/bill_01.jpg .
```

08 Receipt (image ya PDF) edit karne par kya hota hai

Kabhi kabhi AI koi value ghalat parh leta hai. Har user apni receipts, aur admin kisi ki bhi receipt, **Edit Receipt** (ya har card ke chhote Edit button) se theek kar sakta hai. Save dabane par:

- **Original file ko haath nahi lagta.** Wo disk par aur Original scan tab mein saboot ke taur par rehti hai ke dukaan ne asal mein kya chhapa tha.
- **scanned_data** mein AI wali reading rehti hai; **extracted_data** mein corrected values aa jati hain; **vendor_name**, **date**, **total_amount** bhi update hote hain.
- **edited_at** aur **edited_by_id** mein kab aur kis ne likha jata hai - page par "Edited by ... on ..." dikhta hai.
- Ek nayi picture banti hai - **corrected copy** - SireSoft design mein saaf receipt jisme har corrected value, items, totals aur insights hain, aur upar "CORRECTED COPY - edited by ... on ..." likha hota hai. Ye **media/receipts/corrected/receipt-<id>-corrected.png** mein save hoti hai.
- Dobra edit karne par purani corrected copy ki jagah nayi aa jaati hai.
- **Re-scan** original picture ko AI ke paas dobara bhejta hai; nayi reading data ki jagah le leti hai aur edits aur corrected copy hat jate hain.

Database mein ek edit ki hui receipt dikhana

Saari edit ki hui receipts, nayi pehle, kis ne edit ki ke saath:

```
SELECT r.id, r.vendor_name, r.edited_at, u.username AS edited_by, r.corrected_file
FROM receipts_receipt r
LEFT JOIN auth_user u ON u.id = r.edited_by_id
WHERE r.edited_at IS NOT NULL
ORDER BY r.edited_at DESC;
```

Receipt 84 ka pehle aur baad (AI ki value vs corrected value):

```
SELECT scanned_data->'establishment'->>'name' AS ai_name,
       extracted_data->'establishment'->>'name' AS corrected_name,
       scanned_data->'bill_summary'->>'grand_total' AS ai_total,
       extracted_data->'bill_summary'->>'grand_total' AS corrected_total
FROM receipts_receipt
WHERE id = 84;
```

To haan - kisi bhi ek edit ki hui receipt ke liye, chahe wo image ho ya PDF, aap original file, corrected picture, AI ki pehli reading aur corrected data, sab saath saath dekh sakti hain.

09 History page aur kaun kya dekhta hai

	Aam user	Admin (staff)
History mein	Sirf apni receipts	Sab ki receipts, upload karne wale ke naam ke saath
View / Edit / Delete	Apni receipts	Koi bhi receipt
Kisi aur ki receipt ka URL khud likh kar kholna	"Not found" (404) milta hai	Ijazat hai
Admin panel /admin/	Access nahi	Poore database ka read-only view

History mein wo receipts aati hain jo kamyabi se parhi gayin (aur jo re-scan ho rahi hain). Fail hone wali receipts wahan nahi aatin - wo admin panel aur database mein apne error message ke saath nazar aati hain.

10 Presentation ke liye 10 minute ka demo

1. OpenVPN connect karein. **http://10.0.2.2:9001** kholein - app foran khul jaati hai, koi command nahi chalayai. Batayein: ye server par systemd ke neeché port 9001 par chalti hai.
2. Login karein. Upload page dikhayein.
3. Terminal mein: `ssh siresoft@10.0.2.2, phir journalctl -u siresoft-receiptiq-worker -f` taake worker live nazar aaye.
4. 2-3 receipts upload karein (ek photo, ek PDF). Browser mein Queued -> Analyzing -> Done dikhayein jabke worker log har Receipt #N likh raha ho.
5. Ek natija kholein: items, bill summary, insights, original picture.
6. Ek value edit karein (maslan total) aur save karein. "Edited by ...", Corrected copy tab, aur ye ke Original scan nahi badla, dikhayein.
7. Log mein Ctrl+C dabayein. Database kholein: `sudo -u postgres psql -d siresoft_receiptiq_db`.
8. "10 sab se nayi receipts" wali query chalayein - nayi receipts wahan hain. Edit ki hui receipt ke liye ek-receipt aur pehle/baad wali queries chalayein.
9. `ls -lh ~/siresoft-receiptiq/media/receipts/` - upload ki hui files disk par hain.
10. /admin/ -> Receipts khol kar wahi data browser mein dikhayein. Aakhir mein agle page ke faide batayein.

11 Faide (Advantages)

Faida	Kyun ahem hai
Data company ke andar rehta hai	AI model SireSoft ke apne server par chalta hai. Receipts, qeematein aur customer ki maloomat OpenAI, Google ya kisi bahar ki service ko nahi jaatin
Har receipt ka koi kharcha nahi	Na API key, na mahana AI bill - sirf server ki apni bijli. 10 receipts ho ya 10,000, kharcha wahi
Hath se typing ki bachat	Dukaan, tareekh, har item, quantity, qeemat, tax, charges aur totals khud bhar jate hain, chhoti insights ke saath
Kai formats	JPG, PNG, WEBP, GIF aur PDF; ek saath 5 receipts tak
Koi upload zaya nahi hota	AI shuru hone se pehle file save ho jaati hai. AI band ho to receipts line mein intezaar karti hain aur wapas aate hi khud process hoti hain
User ke liye tez	Upload foran ho jata hai; natije ek ek kar ke aate hain. Browser band bhi kar dein to server kaam karta rehta hai
Insaani correction, record ke saath	Ghaltiyan theek ho sakti hain; kis ne aur kab edit kiya, record hota hai; AI ki asal reading muqable ke liye rehti hai
Original saboot hai	Upload ki hui file kabhi nahi badalti. Alag, saaf label wali corrected copy banti hai
Mehfooz access	Login zaroori; user sirf apni receipts dekhta hai; admin sab; files sirf access check ke baad milti hain; passwords hash ki shakal mein save
Hamesha on	systemd server on hote hi app start karta hai aur crash par restart. Kisi ko "chalana" nahi padta
Professional technology	Django, PostgreSQL, Unicorn, systemd - wahi tools jo bari production websites istemal karti hain. Code GitHub par version ke saath
Check karna aasan	Admin panel, SQL queries aur logs se saaf pata chalta hai kya save hua aur worker kya kar raha hai
Aage barhne ki gunjaish	REST API (/api/) maujood hai - baad mein accounting software, Excel export, mobile app ya reports se joda ja sakta hai

12 Kamzoriyan, aur unka hal

Kamzori	Hal
Speed server par munhasir hai - sirf CPU par ek receipt das seconds se le kar kuch minute tak le sakti hai	Natije ek ek kar ke aate hain; GPU ya taqatwar server se bohat tez ho jata hai
AI dhundli, mur-tur ya hath se likhi receipts ghalat parh sakta hai	Edit Receipt se theek karein, ya Re-scan. Hardware ijazat de to bara model (qwen2.5v1:7b) behtar parhta hai
Kai pages wali PDF ka sirf pehla page parha jata hai	Lambi invoices page by page upload karein
10.0.2.2 sirf VPN par chalta hai	VPN ke bahar Cloudflare tunnel ka address istemal karein

Kamzori	Hal
Free Cloudflare address har restart par badal jata hai	journalctl wali command se dobara dekhein, ya pakka domain lagwayein
VPN wale address par connection http hai, https nahi	Traffic VPN ke andar rehta hai; baad mein SSL certificate wala domain lag sakta hai
Sab kuch ek hi server par hai	Database aur media ka regular backup lein (command neeche)

Saada backup (server par chalayein):

```
cd ~

# database ek file mein
sudo -u postgres pg_dump siresoft_receiptiq_db > receiptiq-db-$(date +%F).sql

# saari receipt files ek archive mein
tar czf receiptiq-media-$(date +%F).tar.gz siresoft-receiptiq/media
```

13 Masle aur unka foran hal

Kya nazar aa raha hai	Kya karein
Page nahi khul raha	VPN connected hai? Phir server par: <code>systemctl is-active siresoft-receiptiq</code>
Receipt "Queued" par ruki hai	Worker nahi chal raha: <code>sudo systemctl restart siresoft-receiptiq-worker</code>
Receipt "failed"	Message parhein; <code>journalctl -u siresoft-receiptiq-worker -n 50</code> dekhein; Re-scan karein
Update ke baad tabdeeli nazar nahi aati	<code>collectstatic</code> chalayein aur dono services restart karein; browser mein Ctrl+F5
"Bad Request (400)"	Jo address istemal hua wo <code>.env</code> ke <code>ALLOWED_HOSTS</code> mein nahi hai
Screen ":" ya "(END)" par atak gayi	q dabayein

Poori commands ki list: project mein `docs/COMMANDS.md` aur `deploy/README.md`.